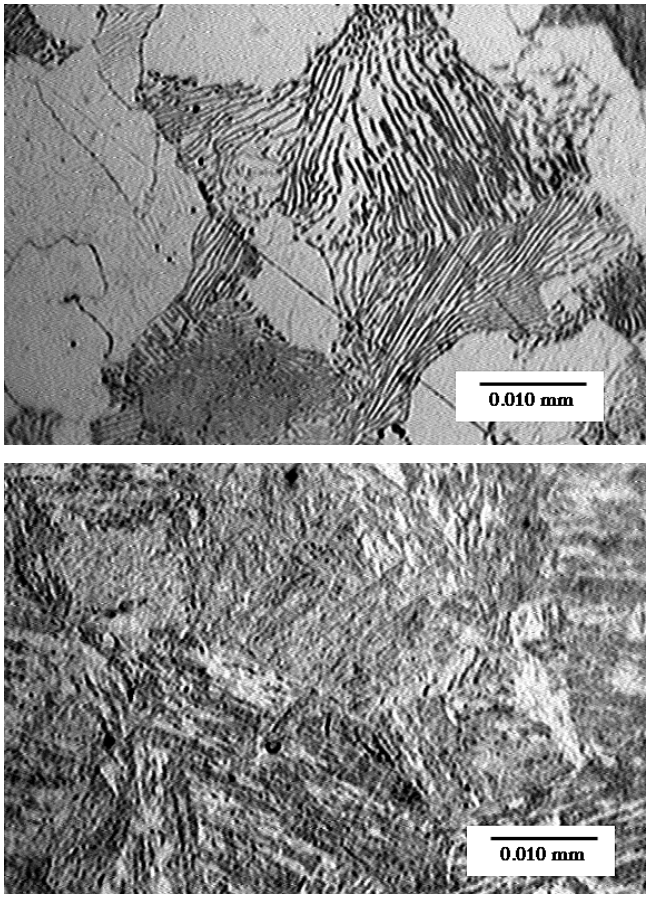
NPRE 432: Engineering Materials Laboratory

Lab 5
(from ME 330)

Heat Treatment of Steel



4340 Steel Micrographs
Annealed (top) and Water Quenched (bottom)

© 2007 The Board of Trustees of the University of Illinois
All Rights Reserved

I. Introduction

An important property of iron is its various solid phases. The transformation into austenite (γ—face centered cubic crystal structure) when heated above the eutectoid temperature, then back into ferrite (α iron—body centered cubic crystal structure) below this temperature allows for a number of microstructural effects to be achieved. The austenite phase is a basic property of iron, making it possible for steel to have a wide range in properties. Alloying is used to affect the process of austenite decomposition into other phases and microstructures.

Since steels contain iron and carbon by definition, ferrite is not the only decomposition product. The solubility of carbon in austenite can be as high as 2.11% (weight percent), while the maximum solubility of carbon in ferrite is 0.022% at 727°C, and less than 0.01% at room temperature. Only a small amount of the carbon in austenite can remain in a solid solution of ferrite. The remaining carbon forms a second phase, iron carbide (Fe_3C), also called cementite. The structure of this phase significantly affects the mechanical properties of the steel. The most common structure an iron – iron carbide mixture forms is pearlite, a complex structure with interleaved planes of ferrite and iron carbide. Additionally, cementite usually forms at the grain boundaries. These equilibrium phases are identified on the Fe – Fe_3C phase diagram in Figure 1.

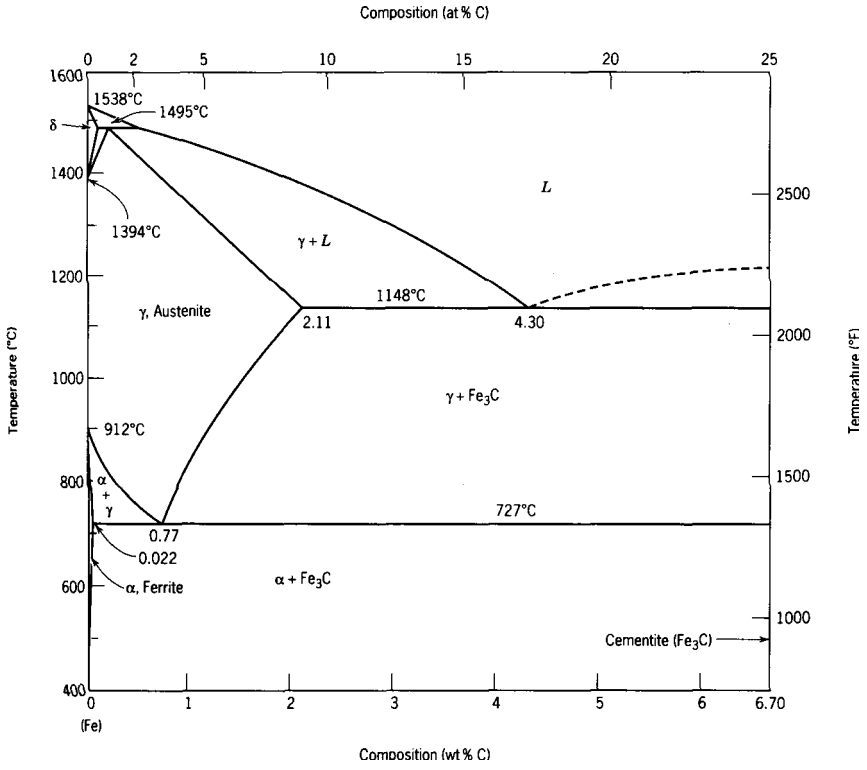


Figure 1: Fe – Fe_3C Phase Diagram

In addition to the equilibrium phases, a variety of metastable phases can also be formed. These include bainite and martensite. Bainite is a mixture of iron carbide in ferrite that gives a strong, tough material, similar to fine pearlite. Pearlite and bainite are considered microstructures since each is composed of two phases – ferrite and iron carbide.

Martensite is a body centered tetragonal crystal structure that is formed during rapid, non-equilibrium cooling in some steels. Martensite is desirable because it is the highest-strength phase possible for iron. Unfortunately, martensite is also quite brittle. Note that martensite does not appear on the iron-iron carbide phase diagram since it is a non-equilibrium phase. Martensite and bainite are metastable phases which mean that the diffusion of carbon through the microstructure continues toward equilibrium even at room temperature, although this transformation may take millions of years to complete. Since martensite is brittle, it is usually tempered to recover ductility at the expense of strength.

The formation of pearlite, bainite, or other structures containing iron carbide cannot occur without the diffusion of carbon atoms through the crystal structure. These reactions are limited by the rate at which carbon atoms can diffuse, hence they are called diffusive reactions. The formation of martensite is not a diffusive reaction since the carbon atoms do not move significantly during the transformation from austenite to martensite. Martensite is formed through a displacive reaction, where each atom is displaced a small distance to allow the new crystal structure (body centered tetragonal) to form. Diffusive reactions depend on both temperature and time, whereas displacive reactions are time-independent, affected only by the temperature.

Steels are usually heat-treated because it is necessary to change the mechanical properties. This is accomplished by changing the microstructure.

The objectives of this laboratory experiment are:

(1) Understand the effects of several of the common heat treatment processes for steel;
(2) Obtain experimental hardness and strength data for steel that has been subjected to different heat treatment processes;
(3) Compare material properties for the different steels and heat treatment processes.
(4) Correlate mechanical properties with microstructural features.

II. Principal Heat Treatment Variables

There are four principal variables during the heat treatment of steel. These are: carbon content, transformation temperature, cooling rate, and alloying elements. These variables affect the heat treatment by changing the temperature and composition at which certain microstructures and phases form.

A. Carbon Content

The diffusion of carbon affects the rate at which austenite can transform to ferrite once cooled below the eutectoid temperature. Recall that carbon is considerably less soluble in ferrite than in austenite. A ferrite grain cannot form from austenite until the local carbon concentration is at or below the solubility of carbon in ferrite. This transformation usually occurs first at the austenite grain boundaries due to the higher energy level associated with the boundary. The rate of formation of ferrite is thus limited by the rate at which carbon atoms can diffuse. As the carbon content of the steel increases, more time is required for the carbon to diffuse and hence, the time for transformation to complete increases. The lever rule can be used in the iron-carbon diagram to predict the fractions of ferrite and iron carbide that will be formed.

B. Transformation Temperature

The temperature of the steel at transformation affects the final products. Cooling slowly through the equilibrium transformation temperature produces large grains and carbide concentrated at the grain boundaries. Larger undercooling prior to transformation provides a greater driving force to form non-equilibrium structures. Undercooling is simply the cooling to a temperature below that required for transformation without the process actually occurring. This phenomenon is due to the high cooling rate and the amount of time required for diffusion (i.e. diffusive and displacive reactions). The temperature at transformation is influenced by the quenchant and the alloy content. The former controls the cooling rate, whereas the latter retards transformation by blocking the diffusion of carbon.

C. Cooling Rate

The iron-carbon phase diagram is an equilibrium diagram, as are all phase diagrams. These diagrams account for thermodynamic effects, explaining the possible reactions and solid solutions. However, they do not account for the effects of the rate at which the reaction occurs. Diffusion kinetics, the rate at which atoms diffuse through the crystal structure, has a large effect on the structures that are formed. The cooling rate influences both equilibrium and non-equilibrium reactions, and thus affects the final microstructure. Information about the products of various cooling rates for a particular individual steel can be found on a time-temperature-transformation (TTT) diagram. The cooling rate also affects the temperatures at which certain reactions occur. Hence, the TTT diagram is also dependent on the cooling rate.

D. Alloying Elements

Due to the mining and manufacturing processes of materials, steel is not pure iron and carbon. Other elements are present, which can be described in two classes: alloys and

impurities. There are elements that are added intentionally to achieve various properties and microstructures. These elements include manganese, chromium, nickel, molybdenum, aluminum, and silicon. The other class of elements is impurities, which include sulfur and phosphorus. The addition of these elements changes the boundaries on the iron-carbon phase diagram, affecting the temperatures and compositions at which various phases and microstructures are observed. A common example of this effect is stainless steel. By adding nickel to steel, the austenite phase is stable at room temperature. One common reason for the addition of alloying elements is to increase the strength of the steel. Thus, 4140 steel (containing chromium and molybdenum) and 4340 steel (containing nickel, chromium, and molybdenum) are considerably stronger than 1040 steel although all three have a carbon content of 0.4%.

In addition to increasing the strength of the steel, many alloying elements contribute additional strength through the heat treatment process by influencing the shape of the TTT diagram. The effects of these alloying elements on the TTT diagram are discussed further in the Hardenability of Steels laboratory.

III. Heat Treatment Processes

There are several heat treatments of interest to engineers. Some of these have been discussed in previous laboratory exercises. In this experiment the processes of annealing, normalizing, quenching, and tempering will be explored.

A. Austenitizing

The first step in the heat treatment of steels is austenitizing, which is heating the steel to dissolve carbon into FCC iron, forming austenite. The general rule for austenitizing is to heat the steel 55°C to 85°C above the upper critical temperature indicated on the phase diagram for the given composition. Sufficient time must elapse for the alloy to completely transform to austenite – typically 30 minutes, plus 1 hour for each 25 mm thickness of the specimen. (The hold time is only necessary to ensure that the transformation to austenite is complete. A simple heat conduction calculation shows that the specimen reaches the desired temperature in a few minutes.) Transformation to austenite is a reaction that requires time to nucleate and grow new crystals. Although the exact temperature for austenitizing varies depending on the alloy, the temperature is usually around 850°C. Different steels have slightly different austenitizing temperatures, so it is customary to check a metals handbook for the proper times and temperatures.

B. Annealing

After the steel has been austenitized, the cooling rate can be carefully controlled to obtain certain microstructures and properties. The process of slow cooling is called annealing, which achieves results near those of equilibrium transformation. Annealing cannot be defined directly in terms of the resulting properties for all materials. Rather, individual materials must be explored to predict the final properties. The most notable effect of annealing is that it produces a large grain size. In steels, annealing produces a coarse pearlite with lower strength and higher ductility compared to normalizing.

Annealing is accomplished by austenitizing, then slowly cooling in a furnace (lab samples cooled at 0.5 ^{o}C/min). The cooling is usually performed by simply turning off the furnace or placing the specimen in another furnace at a lower temperature. If the steel is cooled by transfer between furnaces, the process is sometimes called spheroidizing. This heat treatment is usually used for high-carbon steels and it generates spheroidite, which are sphere-like iron carbide in a ferrite matrix. In spheroidizing, the second furnace temperature is slightly below the transformation temperature. The transformation back to ferrite plus carbide will occur slowly because of the small thermodynamic driving force.

C. Normalizing

If the steel is air-cooled after austenitizing, the heat treatment is called normalizing. This treatment produces a fine pearlite and a uniform, fine-grained microstructure. In general, fine-grained steels are preferable over coarse steels since finer grains tend to have greater strength and toughness. Normalizing results in increased strength and somewhat lower ductility than full annealing.

D. Quenching

The object of quenching is to obtain microstructures that can only be attained through non-equilibrium processes. If the cooling rate from quenching is fast enough, the transformation from austenite to ferrite and iron carbide does not occur and martensite is formed instead. The cooling rate necessary to form martensite depends on a variety of factors, including carbon and alloy content. As expected, the quenching medium has an important effect on the cooling rate. Two quenchants will be considered in this laboratory: water and oil. Water has a greater heat capacity and conducts heat from the surface of the steel faster, and is therefore a more severe quench. The thermal stresses imposed on a component being quenched may be severe enough to cause cracking. If this occurs, a less severe quench may be necessary, such that results from using oil. During the quenching process, the material is quickly transferred to the quenchant

after it is austenitized. Stirring the medium or moving the specimen around rapidly helps to accentuate the effect.

E. Tempering

Martensite has the quality of being exceptionally strong, but it is also brittle. Obviously, a brittle material is undesirable in most design situations. However, it is possible to improve the ductility of martensite by sacrificing some of the strength. Tempering is a toughening process in which the martensite is heated to initiate a ferrite-plus-iron carbide microstructure, which results in tempered martensite. The ductility is improved because the brittle martensite is replaced by a fine dispersion of rigid carbide particles with a tough ferrite matrix. Tempering can also relieve residual stresses generated by the rapid cooling. Tempering heats the steel just enough to allow creep and some limited diffusion to occur, allowing the stresses in the material to relax. Tempering is sometimes called "drawing," which should not be confused with cold-drawing (cold work). Tempering is a function of temperature and time, with a stronger dependence on temperature. It is common to report only the tempering temperature since the process is usually performed for one hour.

IV. Experiments

A. Materials

Three steels, 1018, 1045, and 4340, were subjected to a variety of heat treatments. Tables 1 and 2 summarize strength, ductility, and hardness data for 1018 and 1045 steels that were previously tested. Representative micrographs of 1018 and 1045 steel specimens subjected to heat treatments are shown in Figures A1-A6 in the Appendix. All tempers were performed for one hour.

The experiments in this lab will focus on 4340 steel, investigating the annealed, normalized, oil quenched, water quenched, and tempered heat treatment states.

Table 1: SAE 1018 Cold Rolled Steel Data

Heat Treatment	P_{max} (kN)	d_o (mm)	d_f (mm)	Rockwell Hardness
As Received	22.98	6.34	4.14	R_B = 100.3
Normalized	16.12	6.30	3.65	R_B = 78.0
Annealed	14.86	6.35	4.01	R_B = 67.3
Spheroidized	11.37	6.51	2.71	R_B = 52.1
Water Quenched	46.33	6.35	4.83	R_C = 42.3

Table 2: SAE 1045 Hot Rolled Steel Data

Heat Treatment	P_{max} (kN)	d_o (mm)	d_f (mm)	Rockwell Hardness
As Received	33.18	7.61	5.32	R_B = 93.4
Normalized	31.74	7.62	5.13	R_B = 91.4
Annealed	28.94	7.66	5.41	R_B = 82.6
Spheroidized	20.15	7.42	4.25	R_B = 66.8
Quench Temp: 550°C	43.50	7.53	4.88	R_C = 28.4
Quench Temp: 450°C	54.66	7.55	5.14	R_C = 37.5
Water Quenched	20.15	7.60	7.60	R_C = 57.9

NOTE : ROCKWELL C SCALE FOR QUENCHED SAMPLES

B. Procedures

SAFETY WARNING: A high temperature furnace will be used in this experiment. Serious burns will result if you touch the furnace or heat-treated specimens without protective equipment. Use tongs and heat resistant insulated gloves when handling specimens. A grinder and wire wheel will also be used, which can produce small metal chips that are projected at high velocity. Be sure to wear safety glasses, and follow the procedure demonstrated by your TA or lab staff.

(1) Due to the time involved for annealed heat treatment, these samples will have been heat-treated before class. Note the austenitizing and normalizing temperatures for the alloy under consideration. An individual or group will not test all heat treatments, but remember to enter your data on the test file so all data is accessible.

(2) Samples to be normalized, quenched, and tempered will be in the oven. Remove them with the tongs. Normalized samples are placed on a brick to air cool. Quench the other samples in the appropriate media: either a water quench or special oil bath will be used. *Observe caution—the material coming out of the oven is **red hot**. Wear heat gloves, and wear safety glasses at all times.* Less than 5 seconds should elapse from opening the oven to immersion in the quenching fluid, and the length of the sample should be perpendicular to the fluid surface as it is immersed or the specimen may warp. Some angle while stirring after immersion is acceptable.

(3) Allow approximately 20 seconds for water quench and 2 minutes for the oil quench. Stir the specimen slowly to allow "cool" fluid to contact the metal surface. The end of the specimen gripped by the tongs has a ground notch. It does NOT receive the intended heat treatment (the tongs alter cooling), so measure hardness on the end opposite the notch!

(4) Wash off any excess oil from the samples to be tempered. Place the samples to be tempered into the appropriate tempering furnace. Note the time and set the timer so that you can temper for the correct time period (30 minutes in this lab). These specimens will be removed from the furnace later in the laboratory session.

(5) Use the wire wheel at the grinding bench to remove coarse iron oxide scale from the gage section and grip ends of the heat-treated specimen. Use the two belt sanders to complete specimen preparation. The larger belt sander (wider belt) is used to clean up of the grip end for hardness measurements and allow the tensile grips to “grab” base metal. The smaller belt sander is used to remove scale from the gage section, which will allow accurate measurement of base material gage diameter and minimize extensometer slippage. Use the water in the buckets to cool the sanded portions before touching the specimen; it gets warm enough to burn you.

(6) Obtain the Rockwell hardness for all specimens. Use the grip section **without** the ground notch and record the data in Table 3 so it can be entered in the lab computer during tensile testing. The range of microstructures requires both B and C scale readings, so be sure to note the Rockwell scale that gave a valid reading. Two hardness testers are set for B scale, while the other two have been configured for C scale. Do not change scales on a given machine (other items would also need changing), but use the tester with the appropriate scale and indenter. Remember to use the cylindrical correction factors (appendix of lab 1).

(7) Using digital calipers, measure the gage section diameter of each cylindrical specimen. Use the average of three measurements for each sample to calculate the initial area. Record these and all other required data as listed in Table 3. One measurement of the grip diameter where hardness was measured should be sufficient for the curvature correction.

(8) Tensile tests just like those performed in lab #2 will be conducted for the heat treatments assigned to your group. Load, strain and crosshead deflection data will be collected and

stored in a text data file. Insert the specimen into the tensile grips in the load frame and attach the extensometer. Balance the extensometer, but NOT the load.

(9) Start computer data acquisition/control program and load each specimen at a rate of 2.5 mm/min. Carefully observe the output data and specimen for signs of necking. Continue loading and observing the load output to monitor the final failure load. Record pertinent data from the test as indicated in Table 4.

(10) After fracture occurs, measure and record the final fracture surface dimensions so that a final area can be computed. Remember to input this value into the computer program.

(11) Remove the tempered specimens from the oven. Allow to air cool for at least 5 minutes. Then cool to room temperature using the water bath previously used to quench the specimens. Although technically the specimens should be air cooled to room temperature, 300 seconds usually allows time for all the transformations that will occur (except for some exotic alloys). This shortcut will allow timely completion of the laboratory and avoid singed fingers. Repeat steps 5 through 11 for this tempered specimen.

(12) Use the stereo microscope to observe the appearance of the fracture surface. Sketch these in your notes, noting the portions of "lip" and "bottom" in the cup-cone failures.

(13) Observe the microstructures for all heat treatments of plain carbon steels on the video monitor and the posters. The plain carbon steels have a eutectoid composition of 0.8 wt% carbon and a eutectic temperature of 723 ^{o}C. Microstructural examination of the more exotic alloys such as 4340 where the basic iron-carbon phase diagram will be altered by alloying will be performed in a subsequent lab.

References:

Boyer, H.E., and T.L. Gall (Editors), Metals Handbook: Desk Edition, American Society for Metals, 1985, Section 28.

Callister, W.D., jr., Materials Science and Engineering: An Introduction, Wiley, 1985, Chapter 9.

Flinn, R.A., and P.K. Trojan, Engineering Materials and Their Applications, Third Edition, Houghton Mifflin, 1986, Chapter 6.

Van Vlack, L.H., Elements of Materials Science and Engineering, Fourth Edition, Addison-Wesley, 1980, Chapter 11.

Table 3: Tensile Test Data Sheet

Measurement or Property			Material / Heat Treatment					
Quantity	*Symbol*	*Units*						
Specimen and Fixture Dimensions								
Rockwell Hardness								
Initial Diameter	d_o							
Initial Diameter	d_o							
Initial Diameter	d_o							
Final Diam. or Width	d_f							
Gage Length	L							
Crosshead Loading Rate	$d\delta/dt$							
Calculated Property								
Initial Area	A_o							
Final Area	A_f							
Experimental Quantities								
Proportional Limit	P_y							
Offset yield Load	$P_{0.2\%y}$							
Maximum Load	P_{max}							

DATA FILENAME								

Table 4: Tensile Material Properties

Property			Material / Heat Treatment					
Quantity	*Symbol*	*Units*						
Elastic Modulus	E							
Yield Strength	σ_y							
Offset Yield Strength	$\sigma_{y.2\%}$							
Ultimate Strength	σ_u							
Fracture Stress	σ_f							
Fracture Strain	ε_f							
% Reduction in Area	% RA							
% Elongation	% EL							
Strain Hardening Coef.	K							
Strain Hardening Exp.	n							

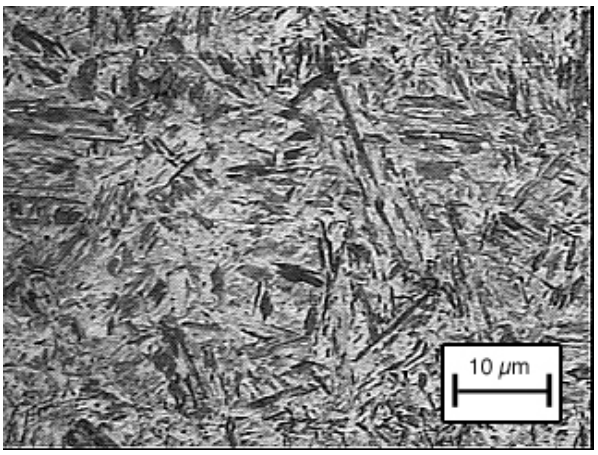


Figure A1: Water Quenched 1045 steel

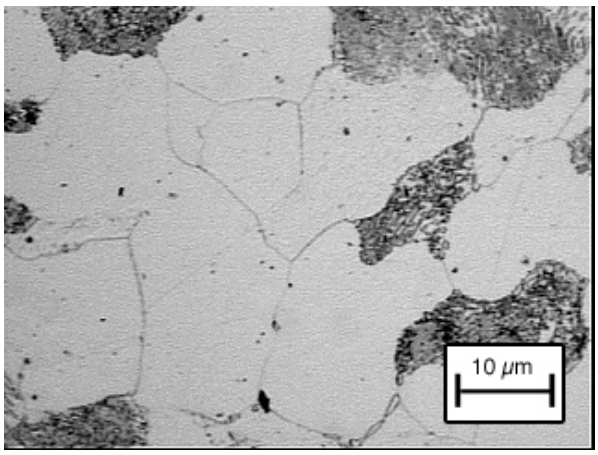


Figure A2: Normalized 1018 steel

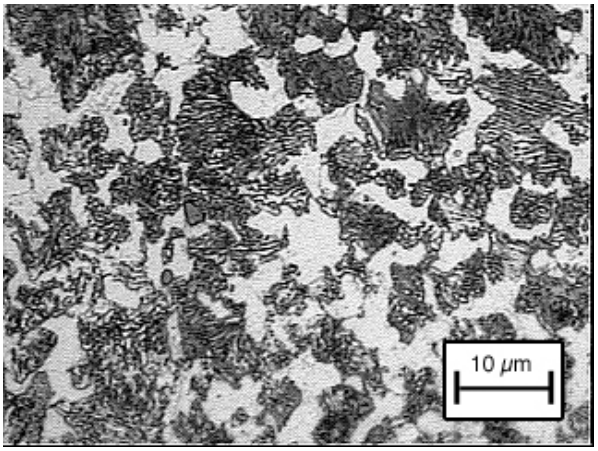


Figure A3: Normalized 1045 steel

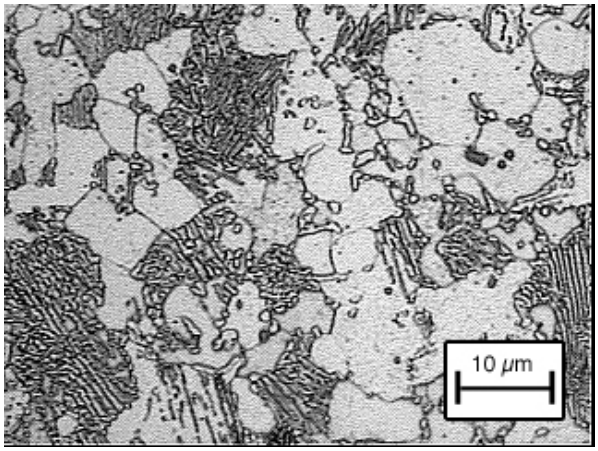


Figure A4: Annealed 1045 steel

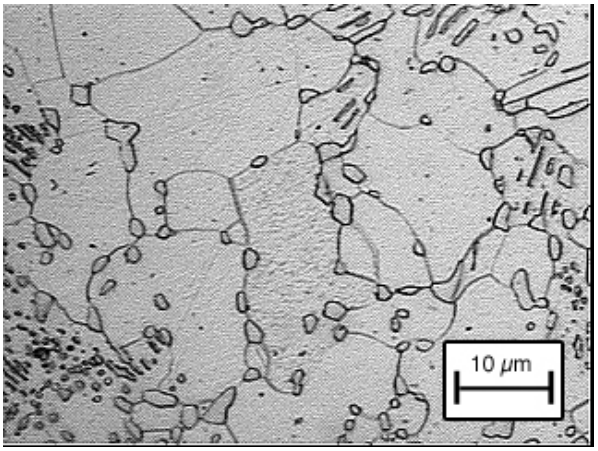


Figure A5: Spheroidized 1045 steel

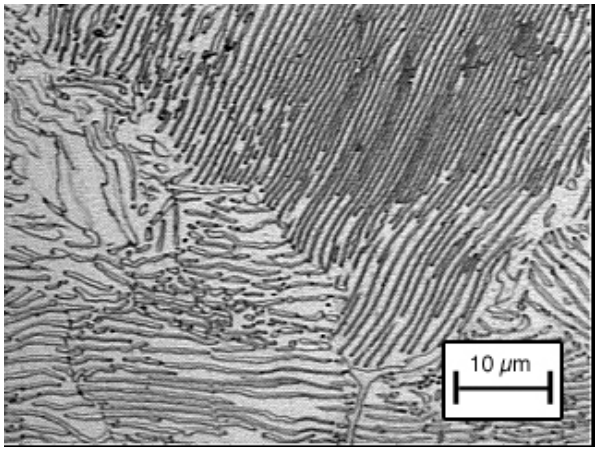


Figure A6: Annealed 1080 steel